ArcGIS Online: Developing with Web Maps and Web Scenes

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ESRI EUROPEAN DEVELOPER & TECHNOLOGY SUMMIT 2026

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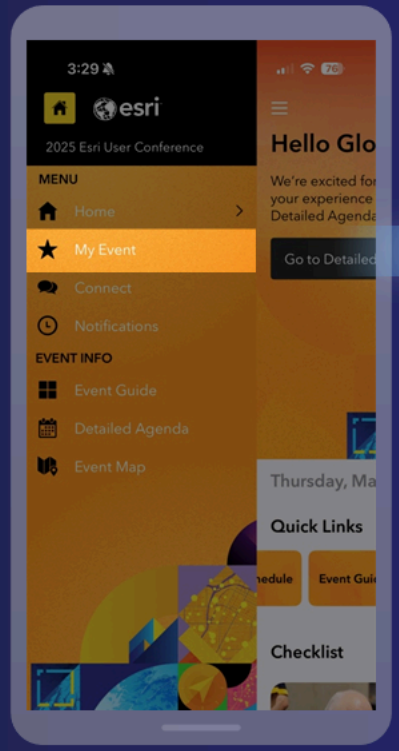

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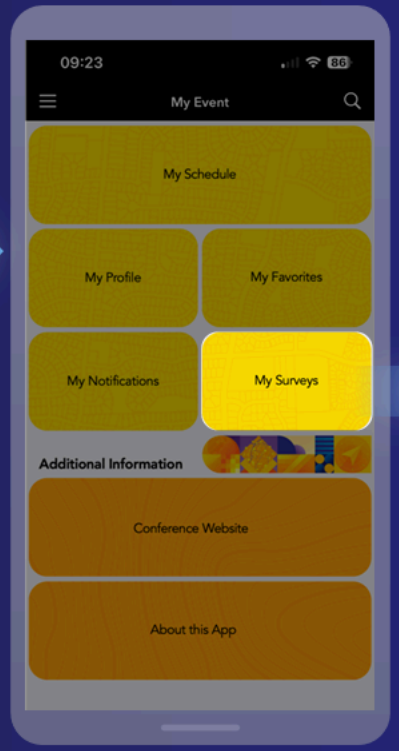

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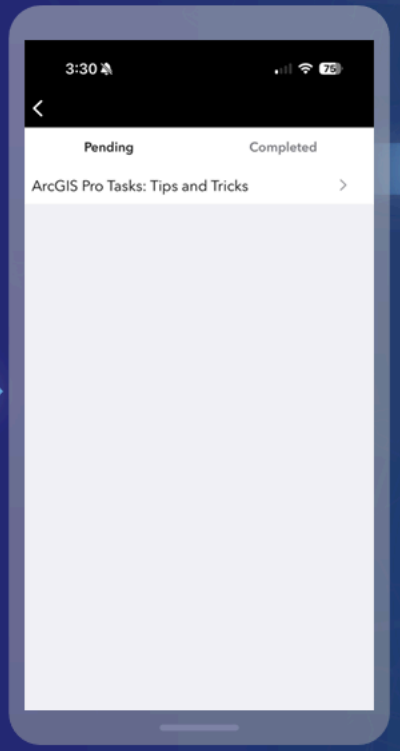

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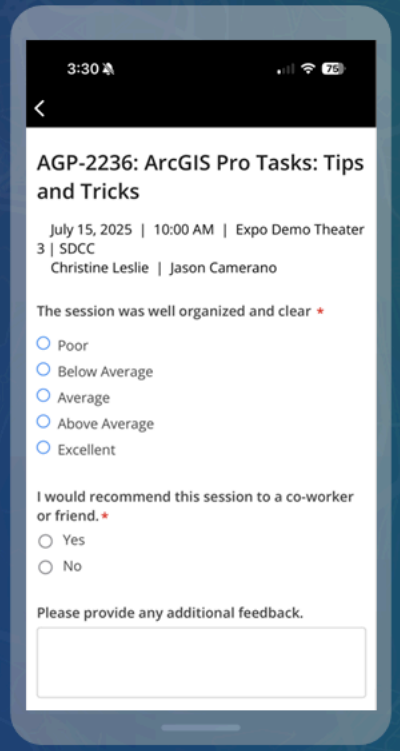
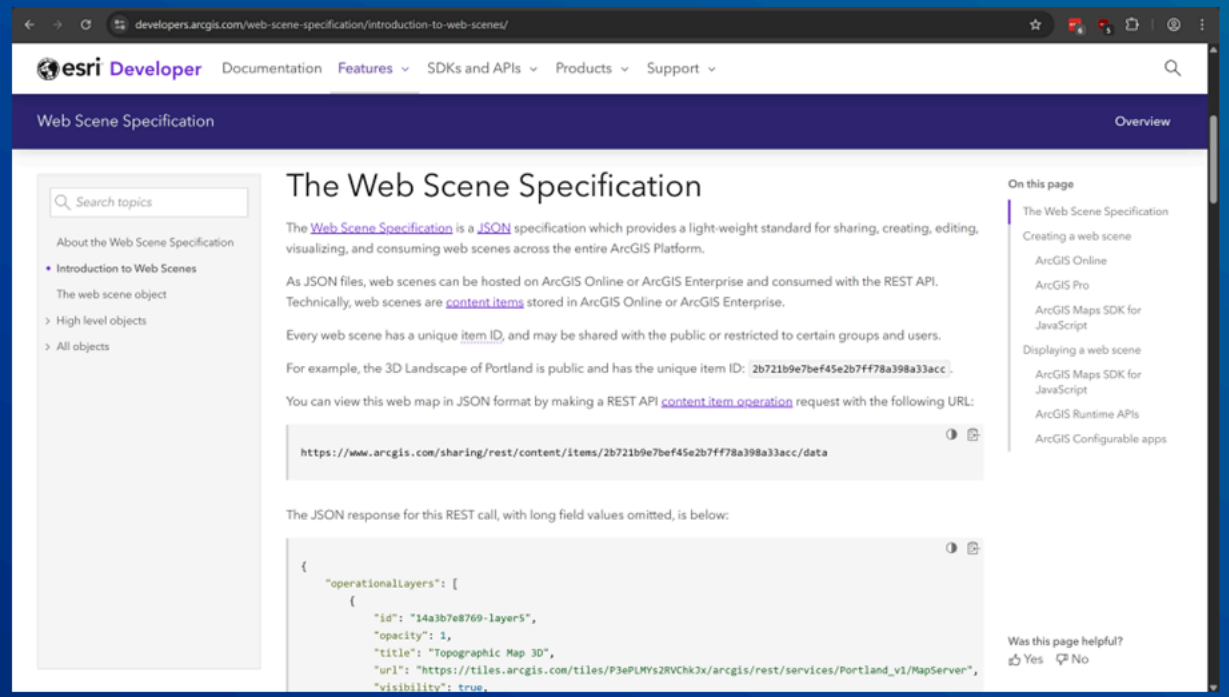


Outline

1. Core concepts
2. Map authoring and consumption
3. Building interactive apps from Web Maps and Web Scenes
4. Authorization and permissions
5. Closing remarks and Q&A

Core Concepts

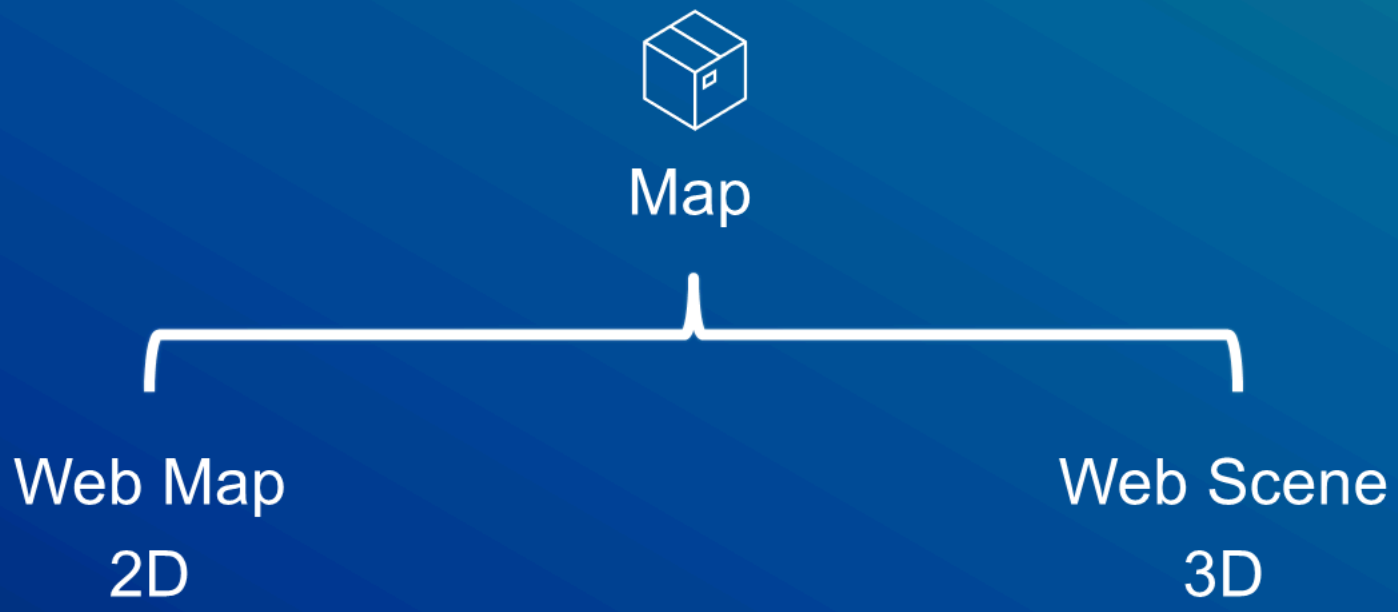
What is a WebScene/WebMap



https://arcg.is/1XTv0S2

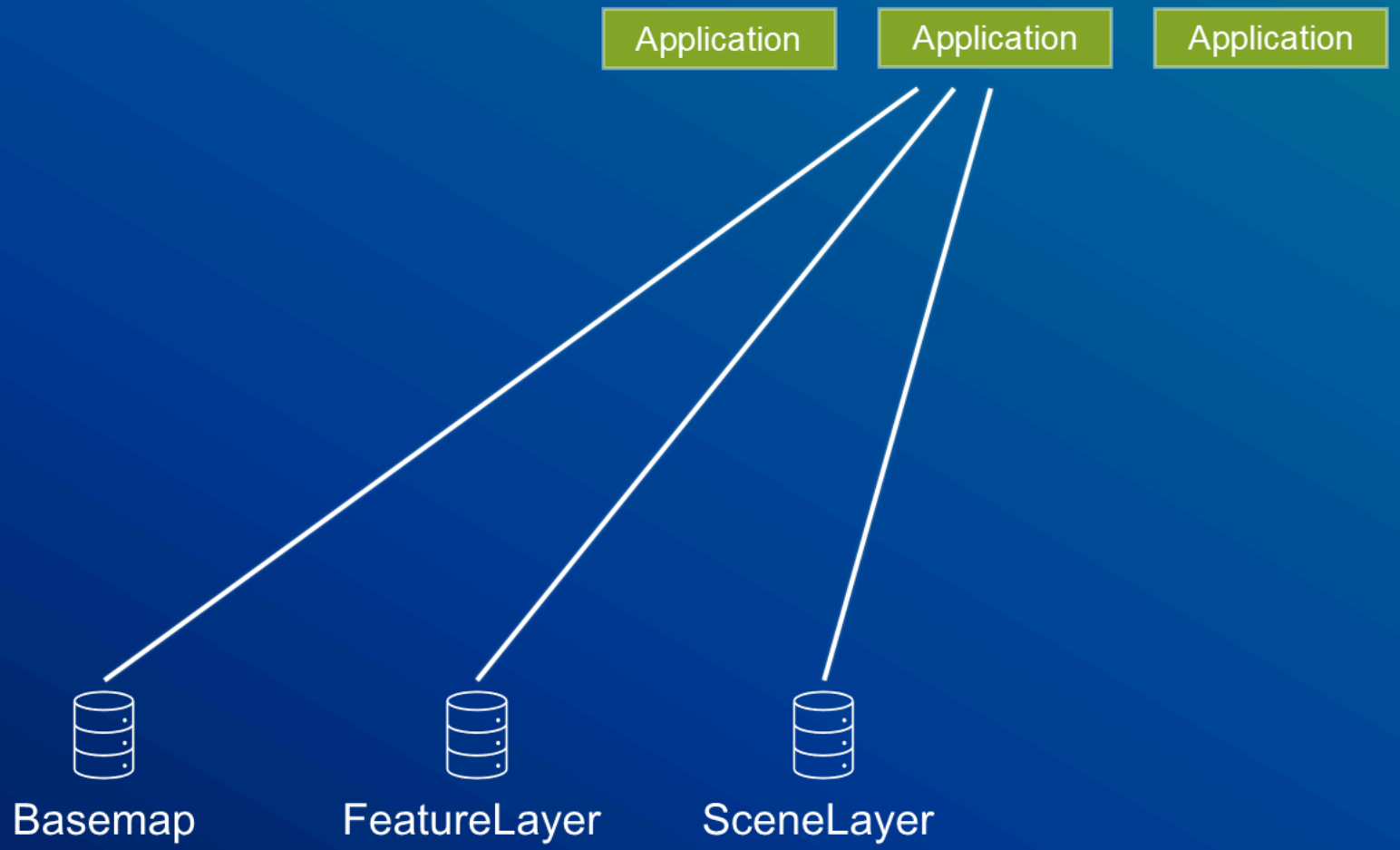
```
{
    "operationalLayers": [ ... ],
    "baseMap": {
        "baseMapLayers": [ ... ],
        "id": "basemap",
        "title": "Topographic",
        "elevationLayers": [...]
    },
    "ground": {
        "layers": [...]
    },
    "heightModelInfo": ...,
    "version": "1.10",
    "authoringApp": "WebSceneViewer",
    "authoringAppVersion": "6.1.0.0",
    "presentation": ...,
    "initialState": {
        "environment": ...,
        "viewpoint": ...
    },
    "spatialReference": ...,
    "viewingMode": "global"
}
```

Web Map vs. Web Scene



Web Apps in ArcGIS

Bigger picture

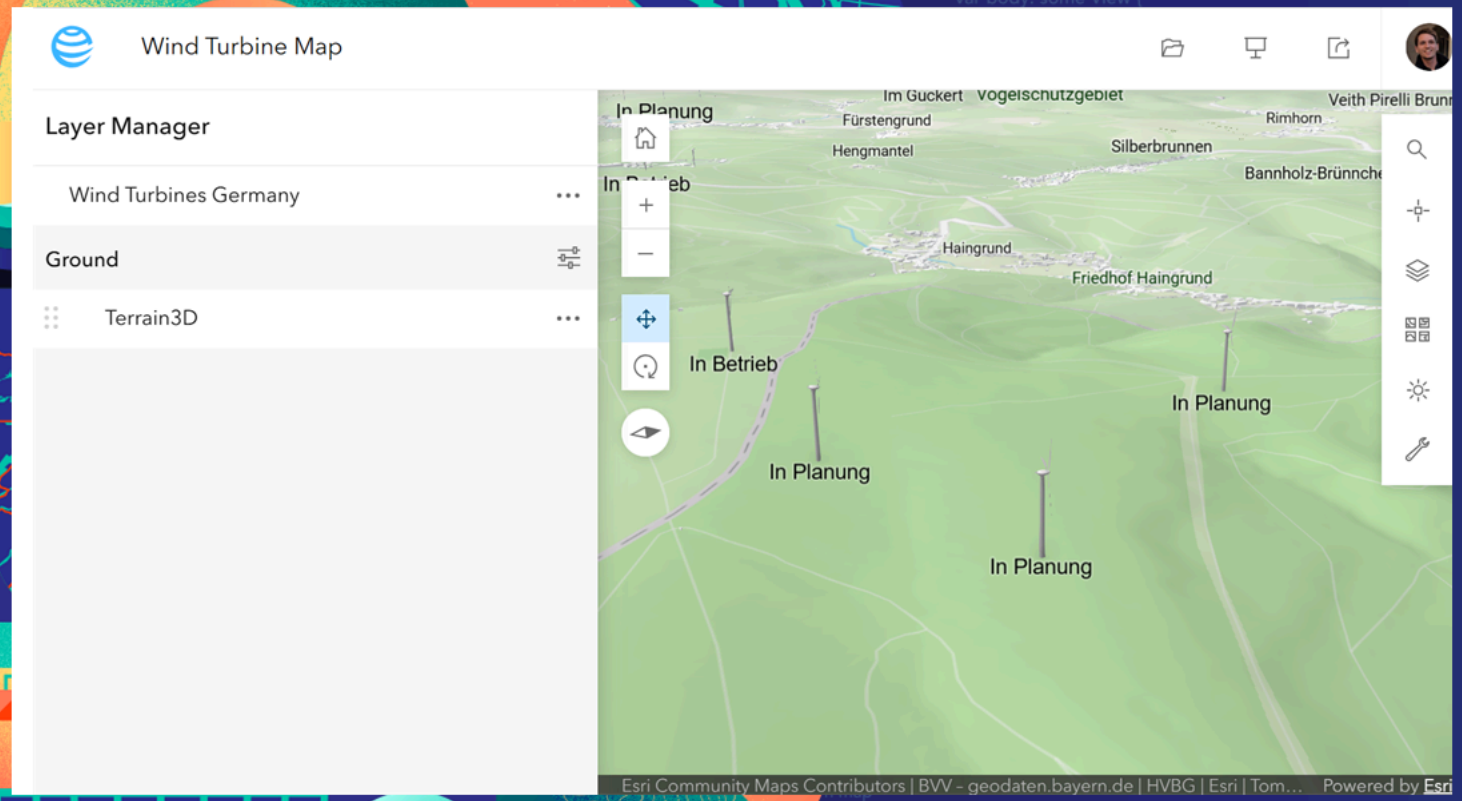


Purpose of Web Maps and Web Scenes

Outline

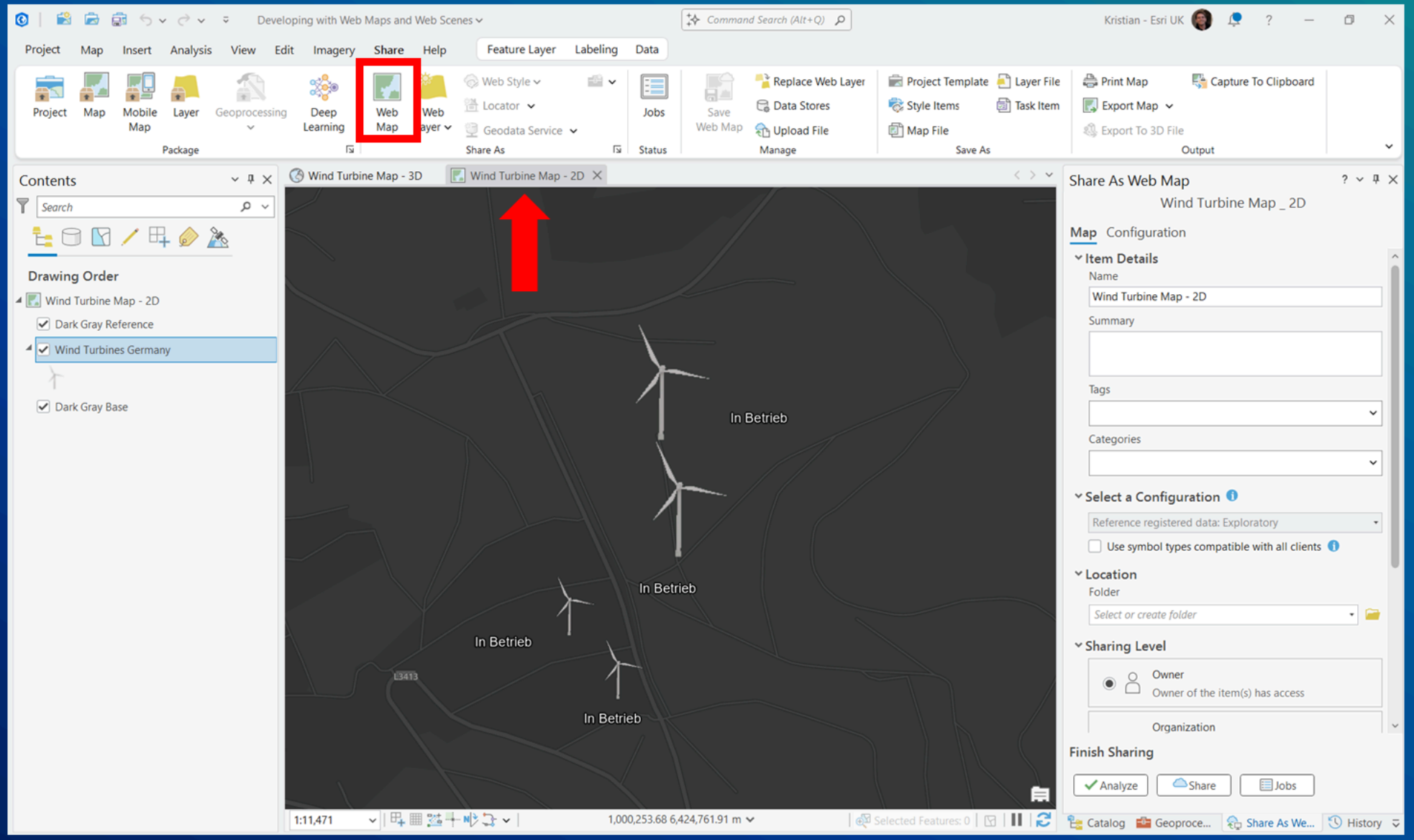
1. Core concepts
2. Map authoring and consumption

Map authoring and consumption



Authoring via ArcGIS Online

Authoring via ArcGIS Pro

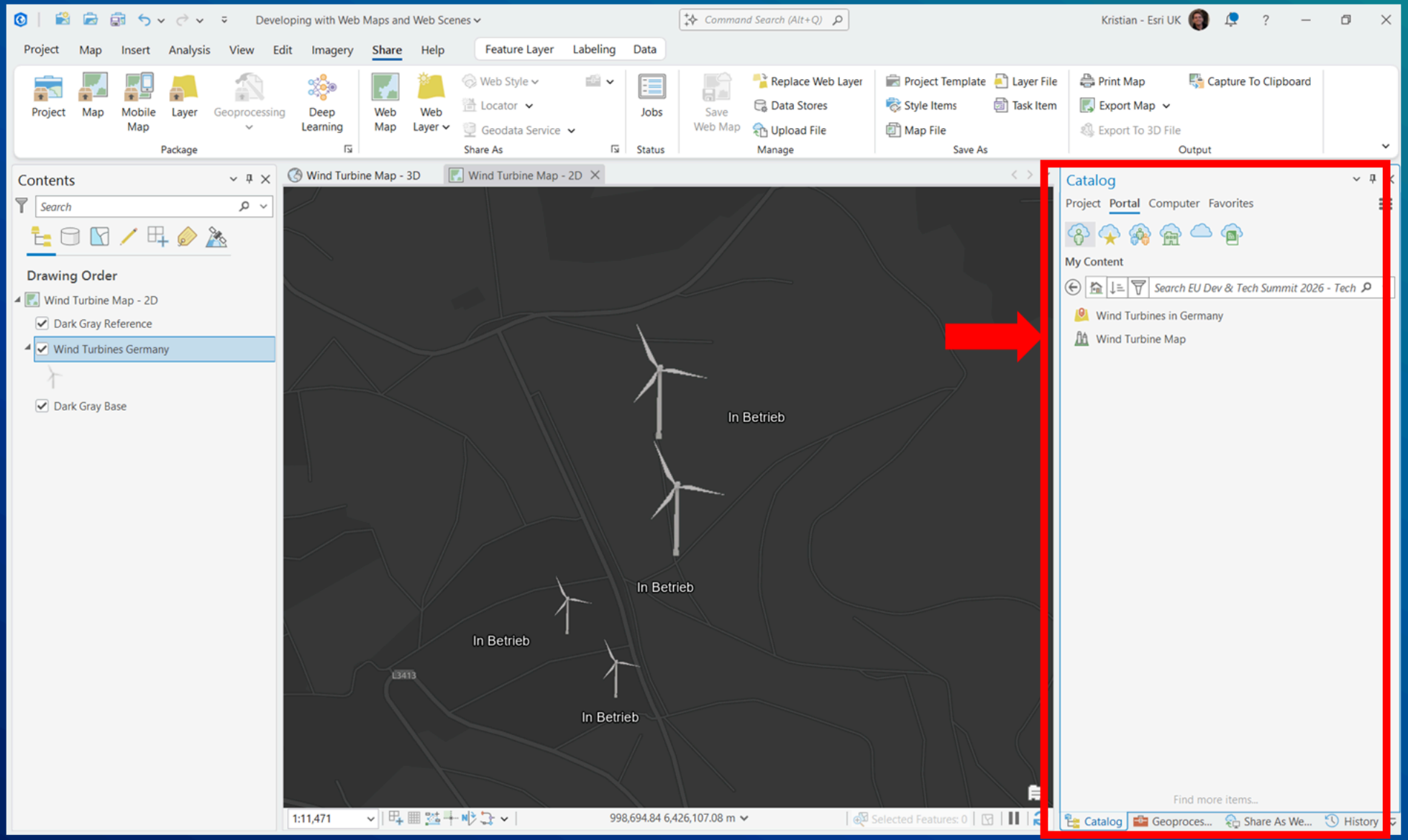


Authoring via ArcGIS Maps SDK for JavaScript

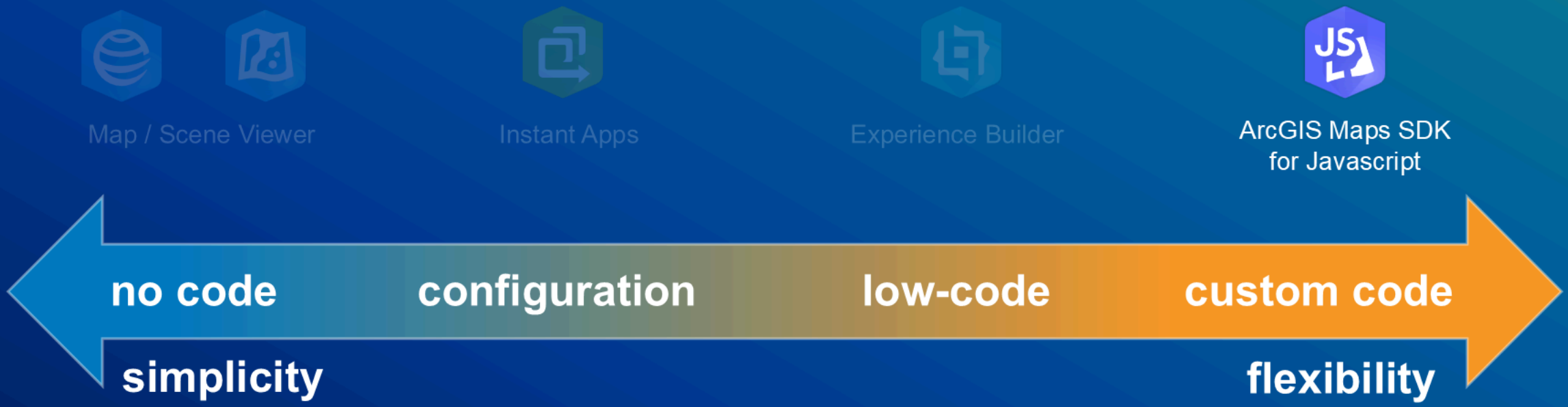
```
await map.updateFrom(view);

const portalUrl = "https://zurich.maps.arcgis.com/";
const item = await map.saveAs({
  portal: {
      url: portalUrl
    },
    title: "forest area with trees"
  },
  {
    folder: "87bb8b00a05e46ab8bad08fec83c241e"
  });

const link = document.getElementById("open-webscene-link");
link.href=`${portalUrl}/home/webscene/viewer.html?webscene=${item.id}`;
const alert = document.getElementById("webscene-saved-alert");
alert.open = true;
```

Consumption of Web Maps and Scenes

Consumption – ArcGIS Pro



Consumption – ArcGIS Online



Consumption – ArcGIS Online – Maps SDK for JavaScript

Maps SDK for JavaScript

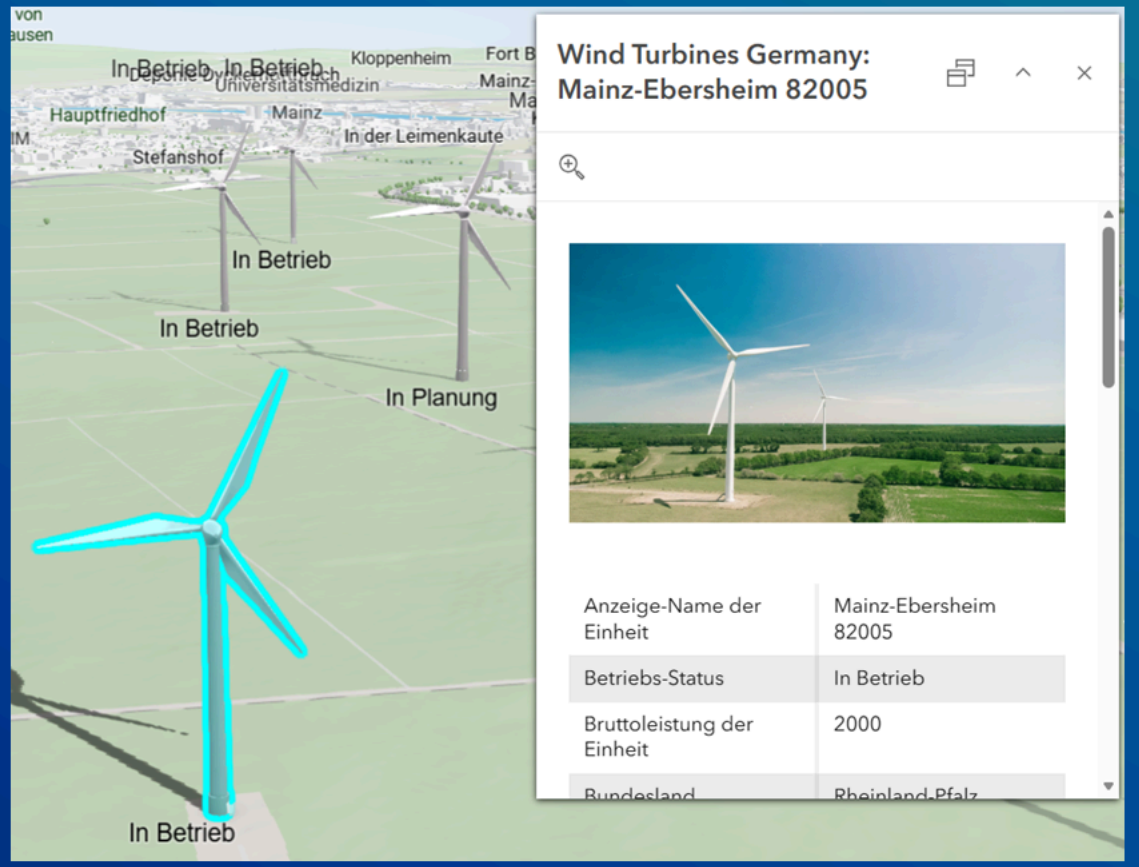
Outline

1. Core concepts
2. Map authoring and consumption
3. Building interactive apps from Web Maps and Web Scenes

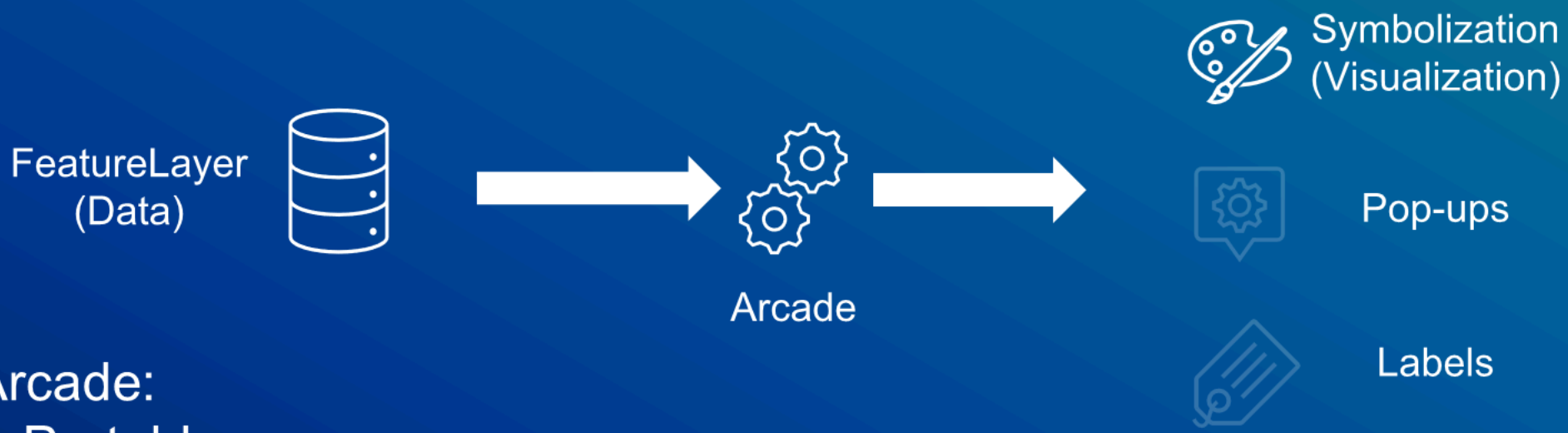
Building interactive apps from Web Maps and Web Scenes

Popups

```
view.openPopup({
  features: [...]
});
```



Arcade – Expression language for WebGIS



Arcade:

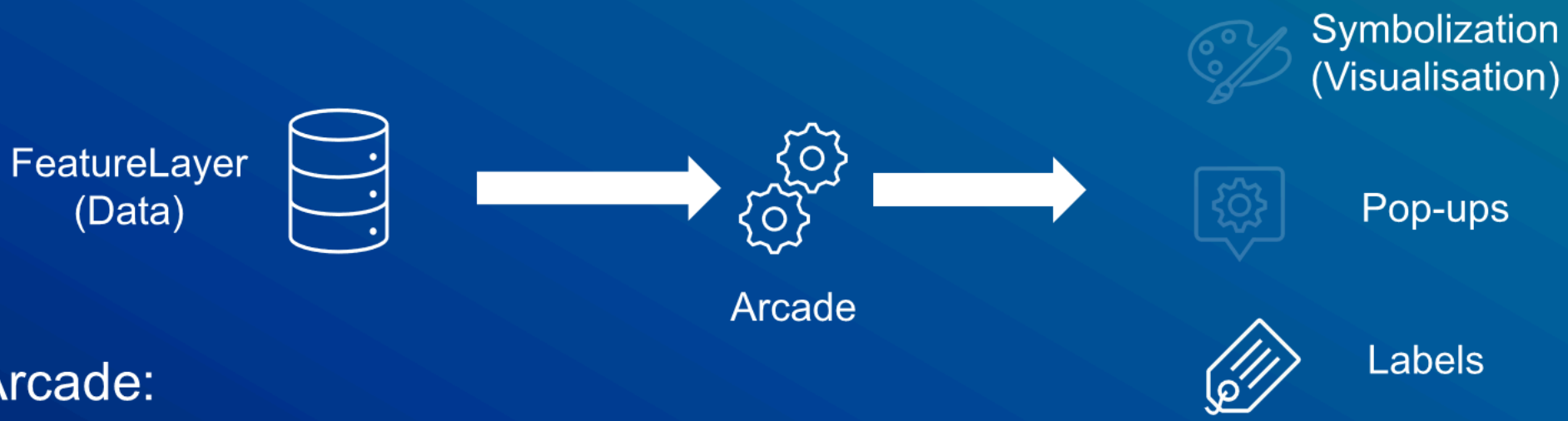
- Portable
- Secure
- Lightweight and fast
- Geospatial built-in

Wind Turbine height

```
/**
 * Converts a German-formatted number string to an Arcade number.
 * Removes thousands separators and replaces the decimal comma with a decimal point.
 *
 * @param {String} numberString - A German-formatted number string, e.g. "1.234,56".
 * @returns {Number} The converted number, e.g. 1234.56.
 */
function convertNumber(numberString) {
  var noThousandSeparator = Replace(numberString, '.','')
  var swapDecimalSymbol = Replace(noThousandSeparator, ',','.')

  return Number(swapDecimalSymbol)
}

return Round(
  convertNumber($feature.Nabenhöhe_der_Windenergieanlage) +
  convertNumber($feature.Rotordurchmesser_der_Windenergieanlage) / 2
  ,2
)
```

Visualisation with Arcade

Arcade – Expression language for WebGIS



Arcade:

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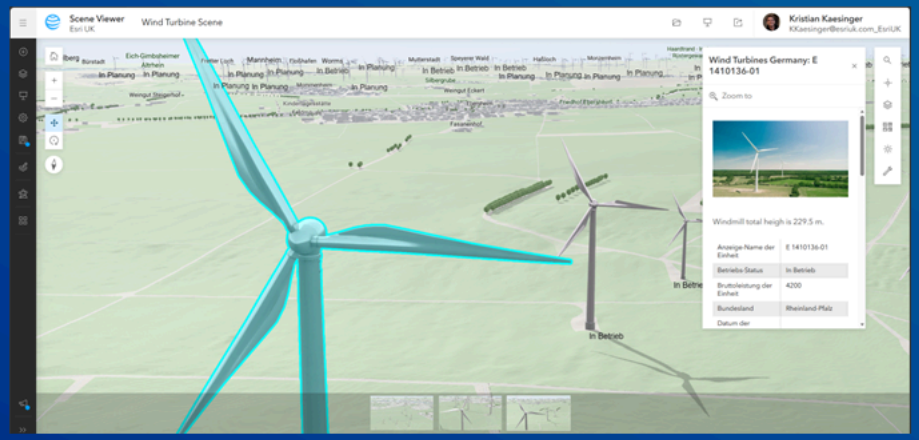
- Geospatial built-in

Labelling with Arcade

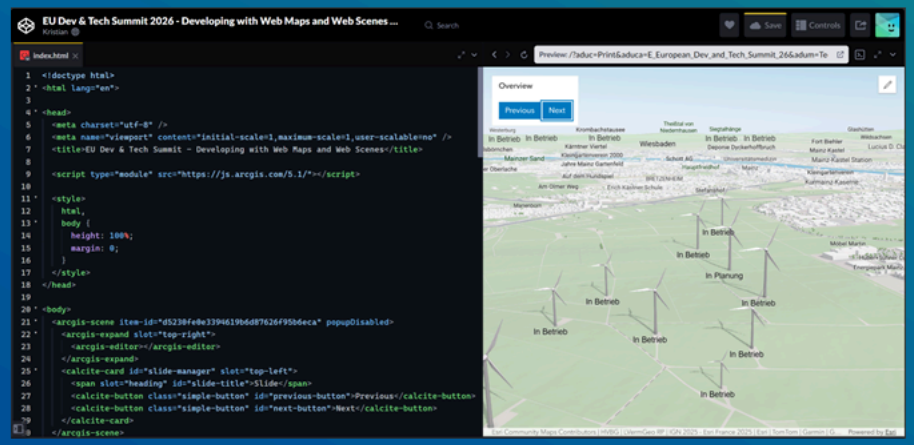
Presentations

WebScene



https://link.esri.com/26eurodevtechsessions/AGO-1097-web-scene-example?qr=true

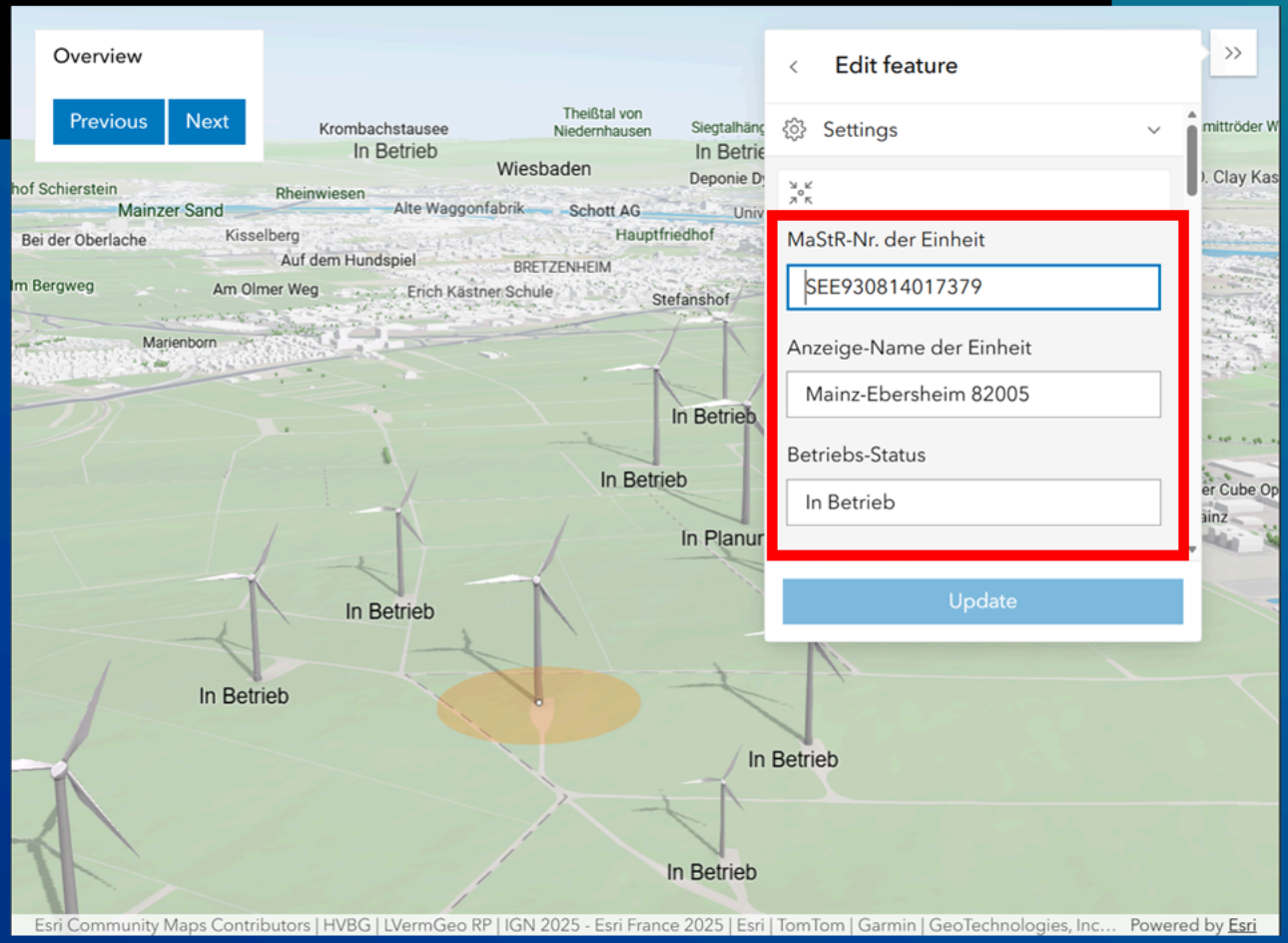
App



https://link.esri.com/26eurodevtechsessions/AGO-1097-codepen-resources?qr=true

Components

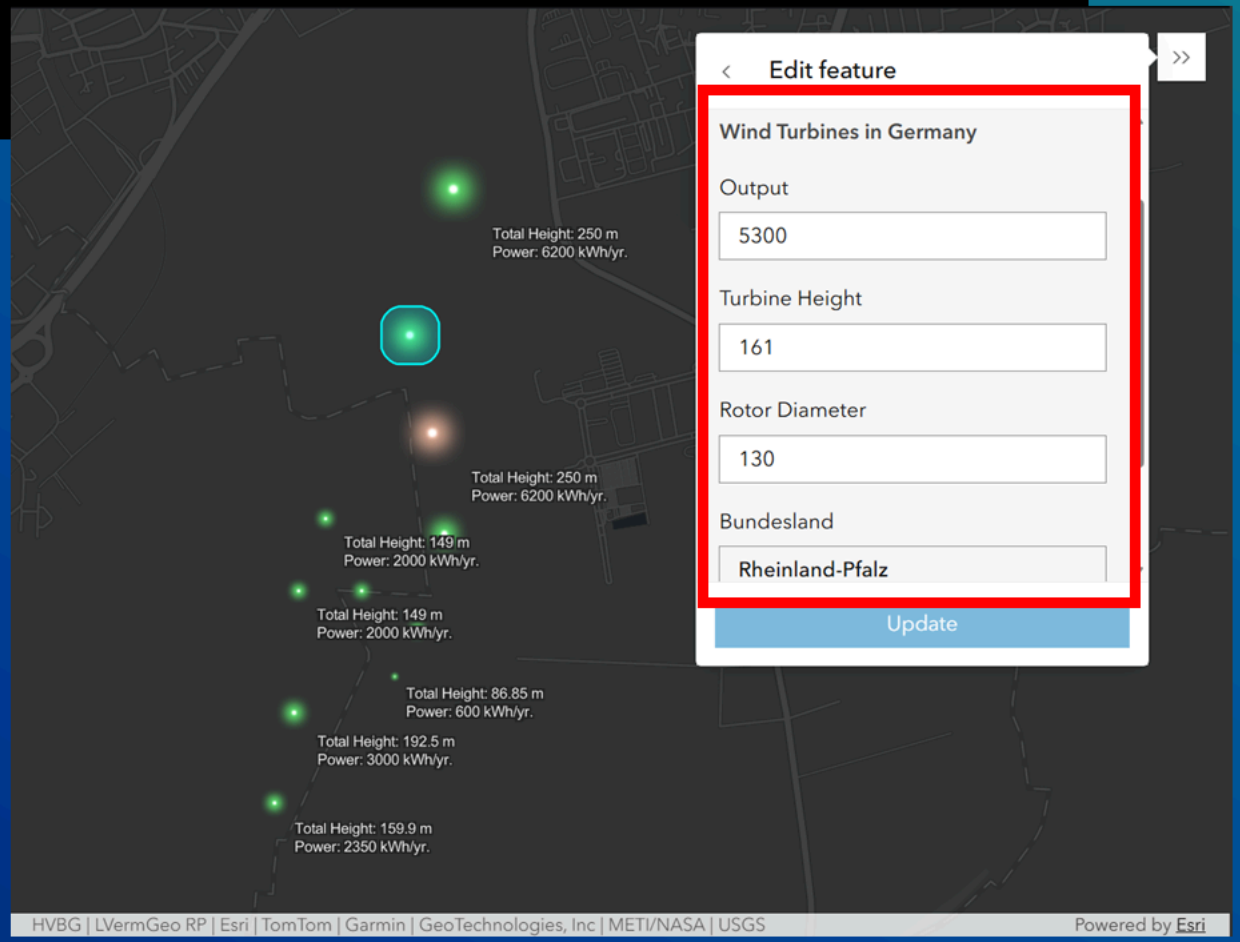
```
<arcgis-scene item-id="67d5fad755d34176b0d7aa06ad0a3f6c" popupDisabled="true">
  <arcgis-expand slot="top-left">
    <arcgis-editor />
  </arcgis-expand>
</arcgis-scene>
```



Feature forms

ⓘ Web Map only

```
<arcgis-map item-id="67d5fad755d34176b0d7aa06ad0a3f6c">
  <arcgis-expand slot="top-right">
    <arcgis-editor />
  </arcgis-expand>
</arcgis-map>
```



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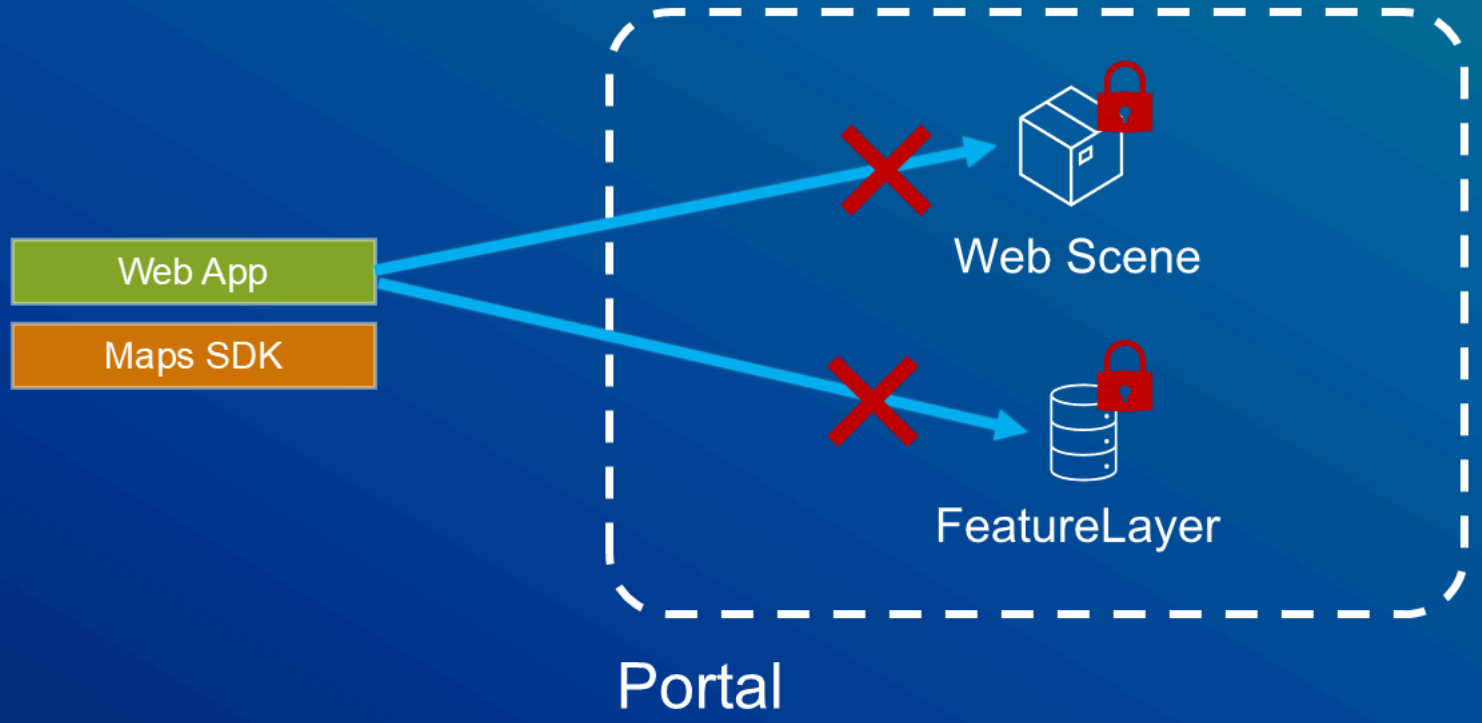
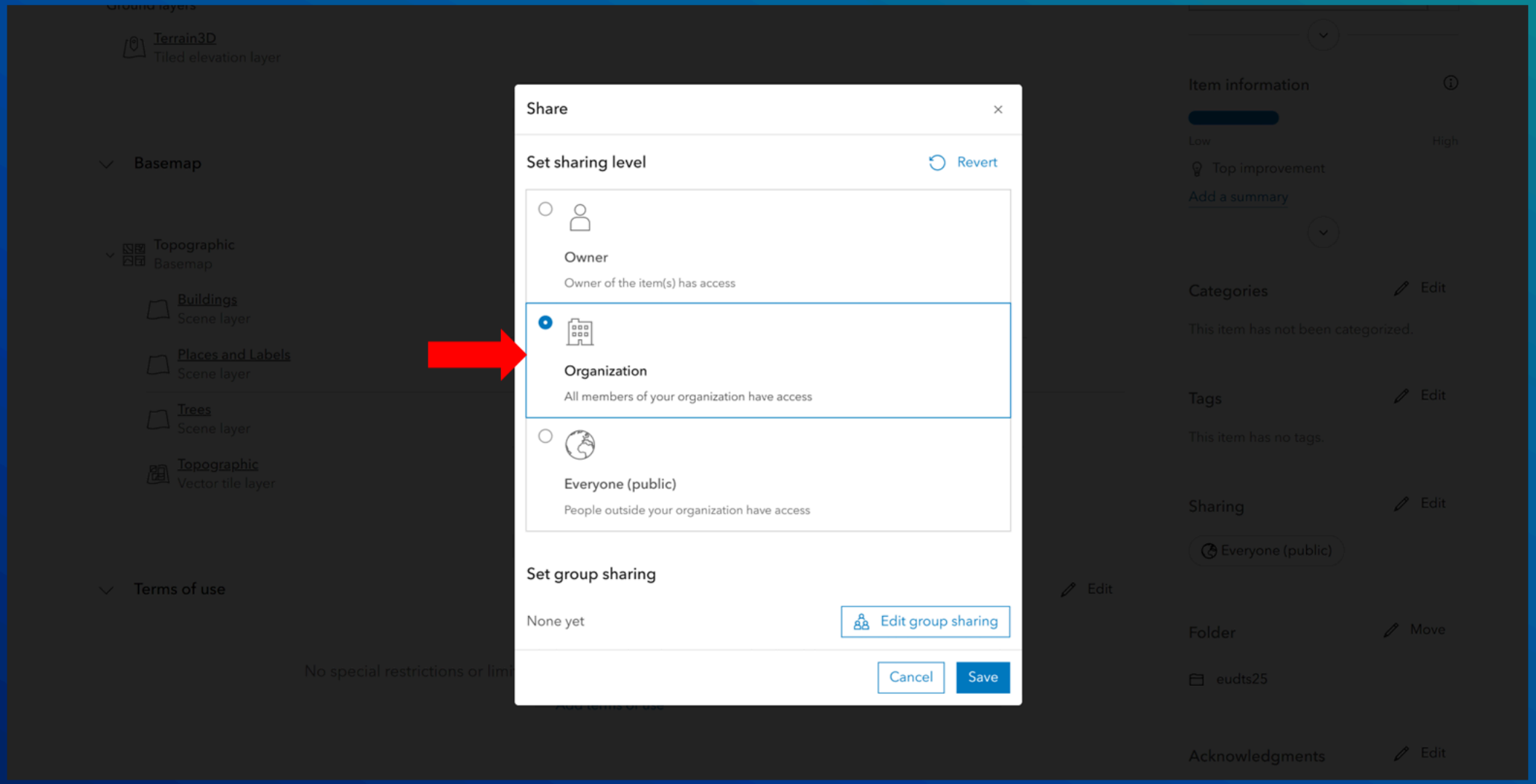
Authorization and Permissions

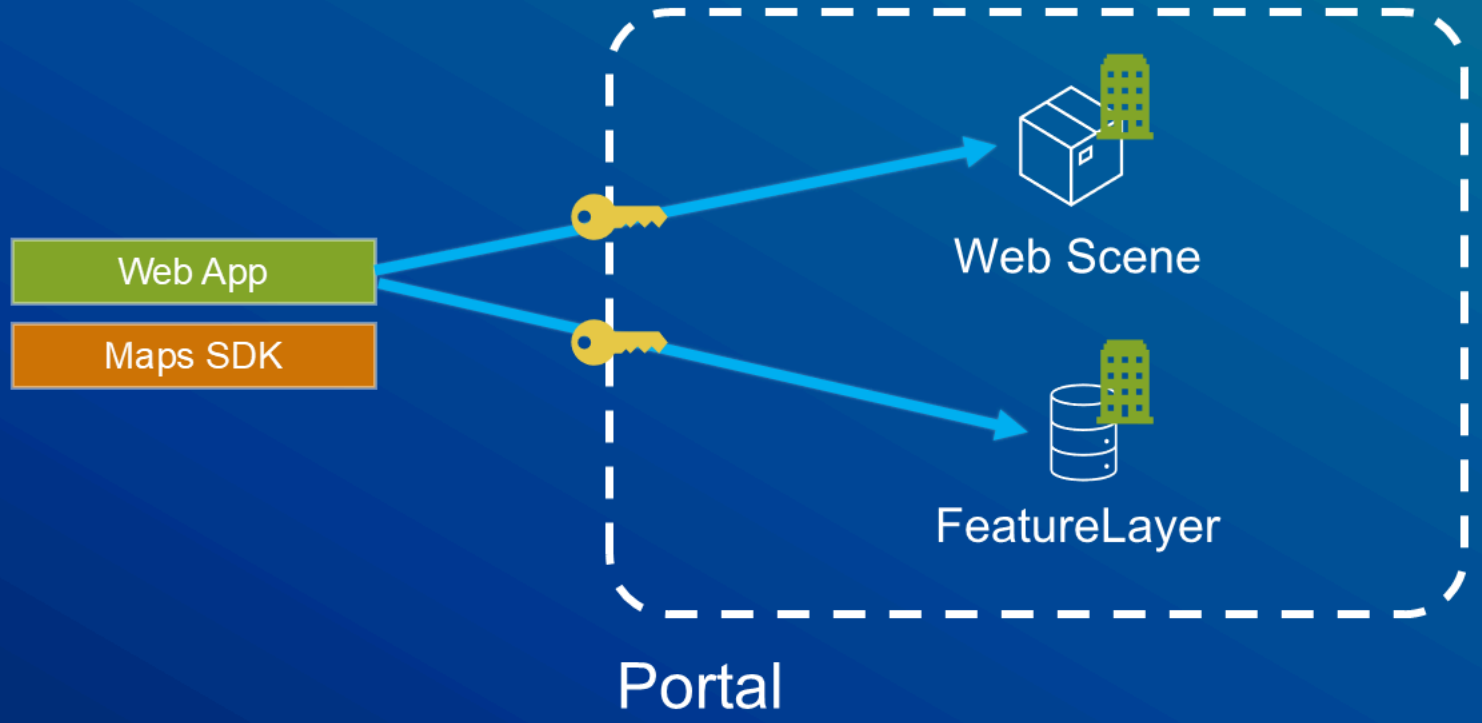
Authorization – Portal item access



Authorization – Sharing



Authorization



Authorization – App

User Authentication

Login screen

Billed towards **user**

Privileges defined by **user**

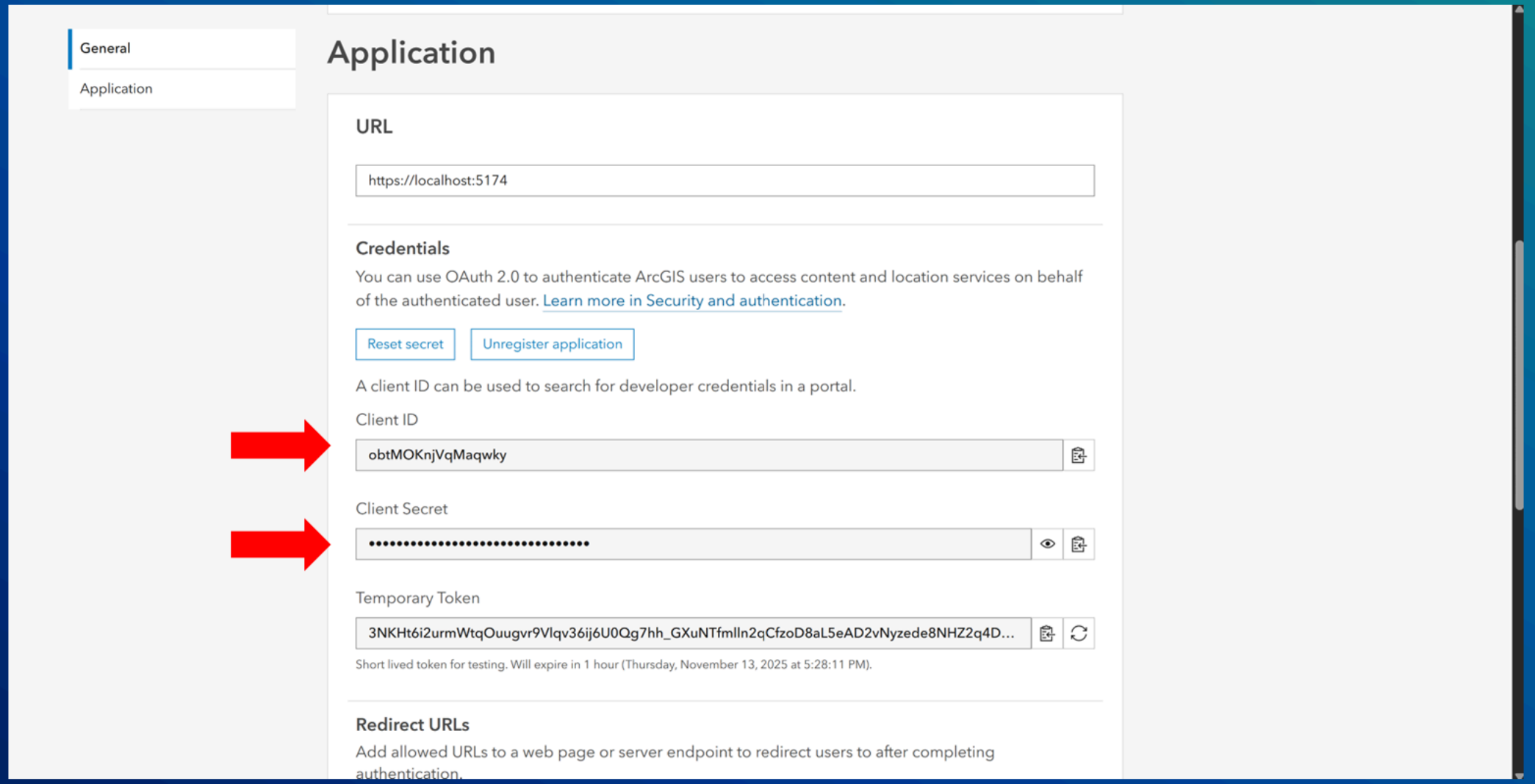
App Authentication

No login screen

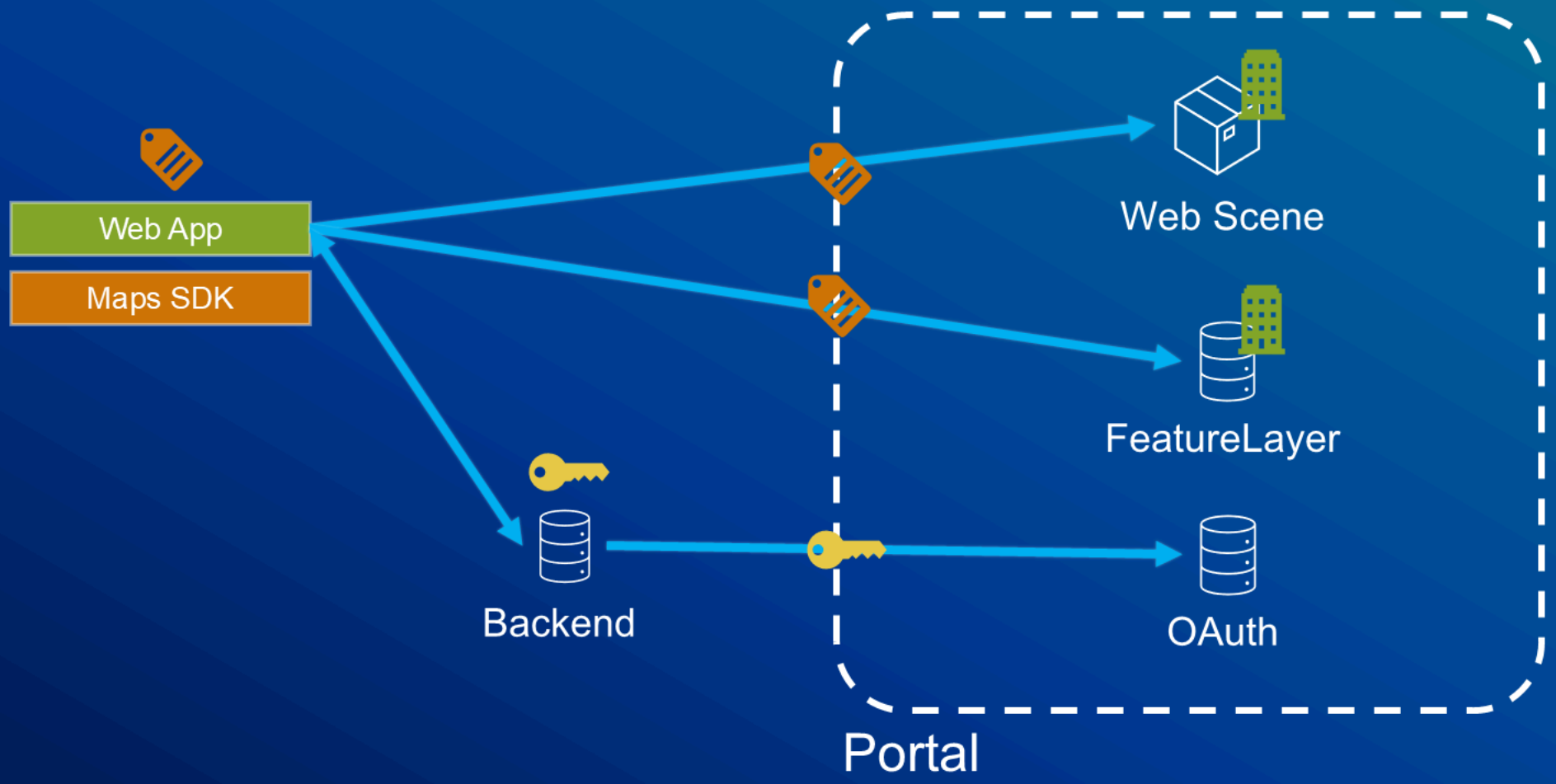
Billed towards **developer / organisation**

Privileges defined by **credentials item**

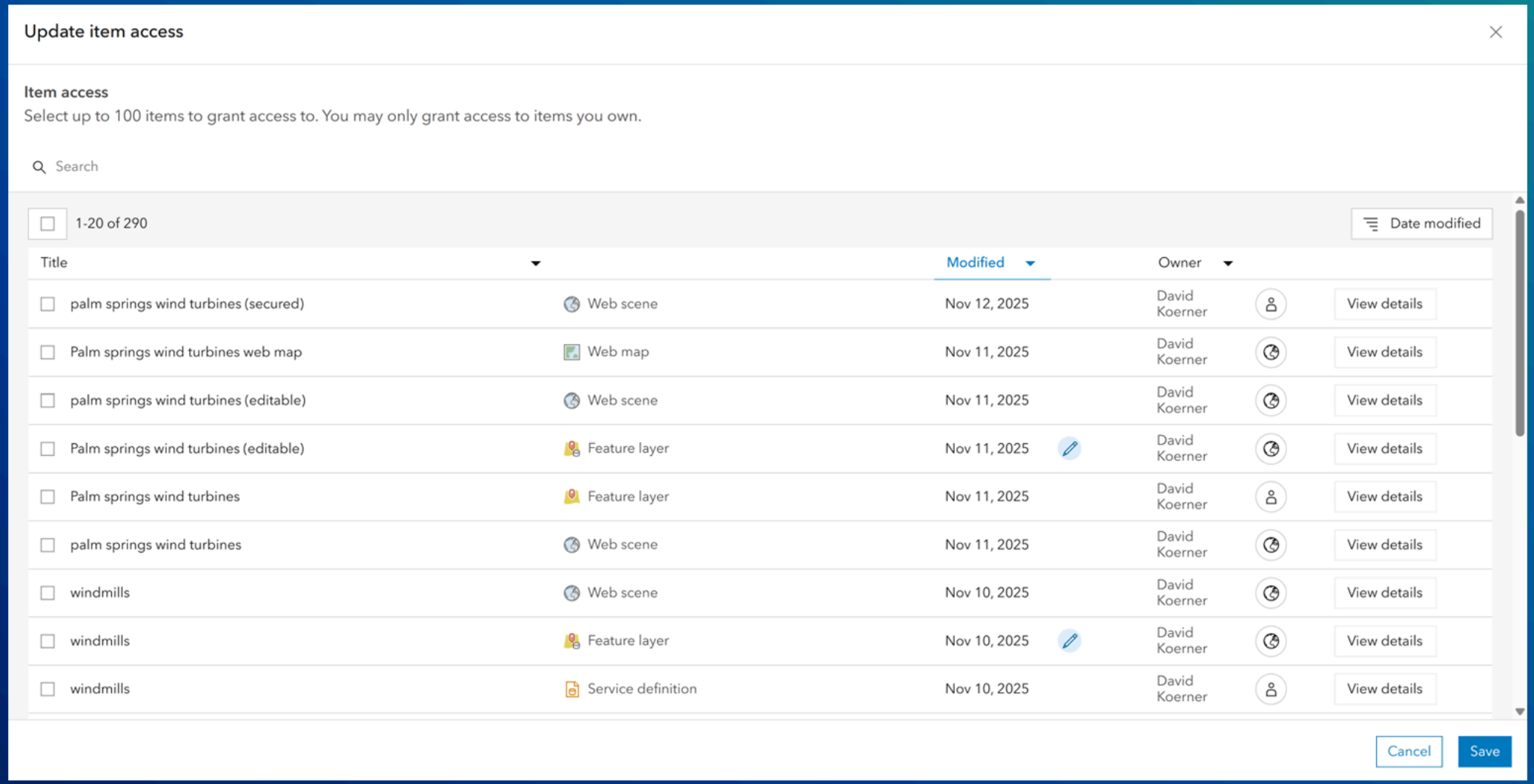
Authorization – Portal item access



Authorization – Oauth



Authorization – Portal item access



Authorization

API Key Authentication

No login screen

Billed towards **developer**

Privileges defined by **credentials item**

User Authentication

Login screen

Billed towards **user**

Privileges defined by **user**

OAuth 2.0

App Authentication

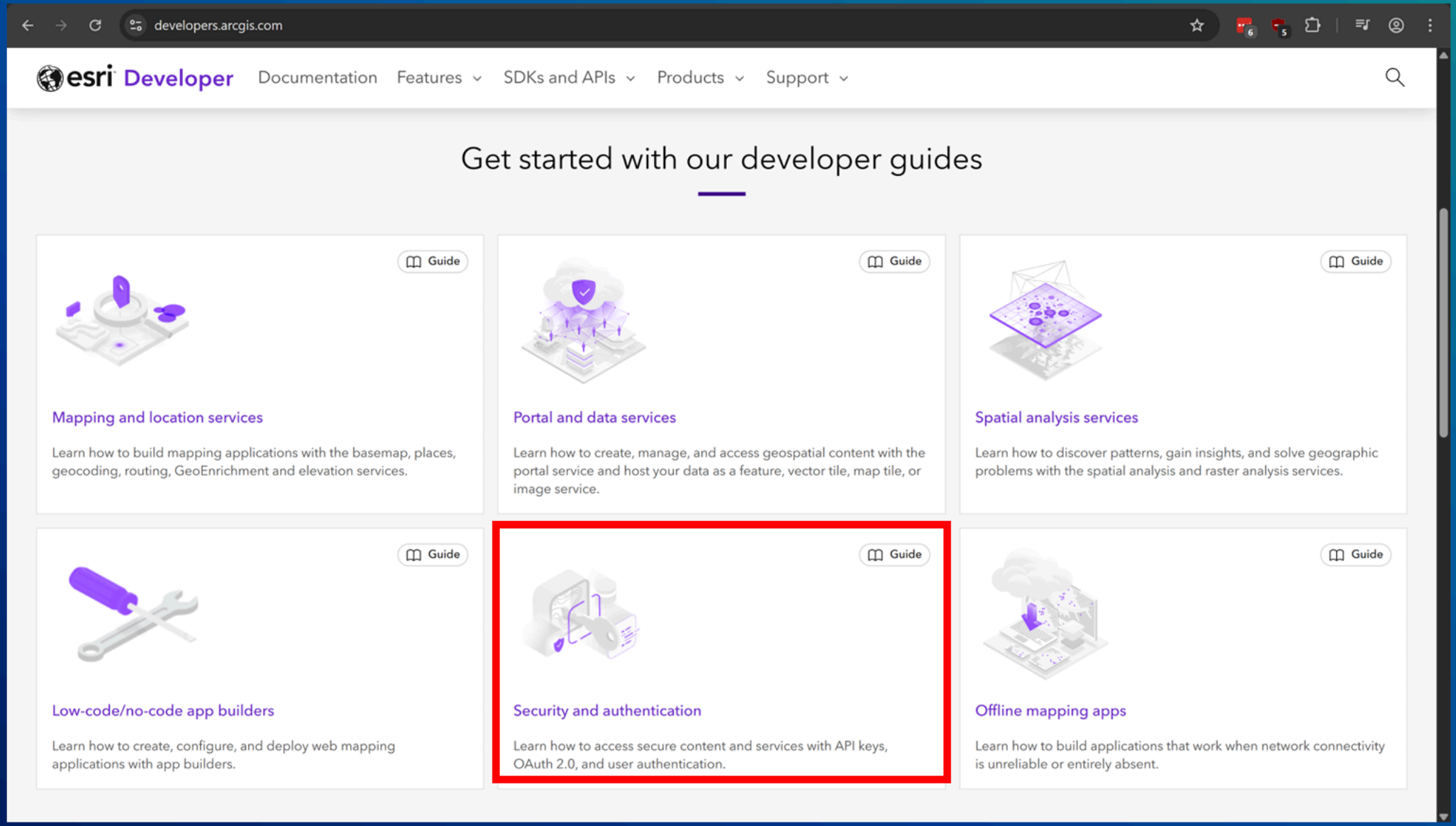
No login screen

Billed towards **developer**

Privileges defined by **credentials item**

OAuth 2.0

Authorization – Oauth



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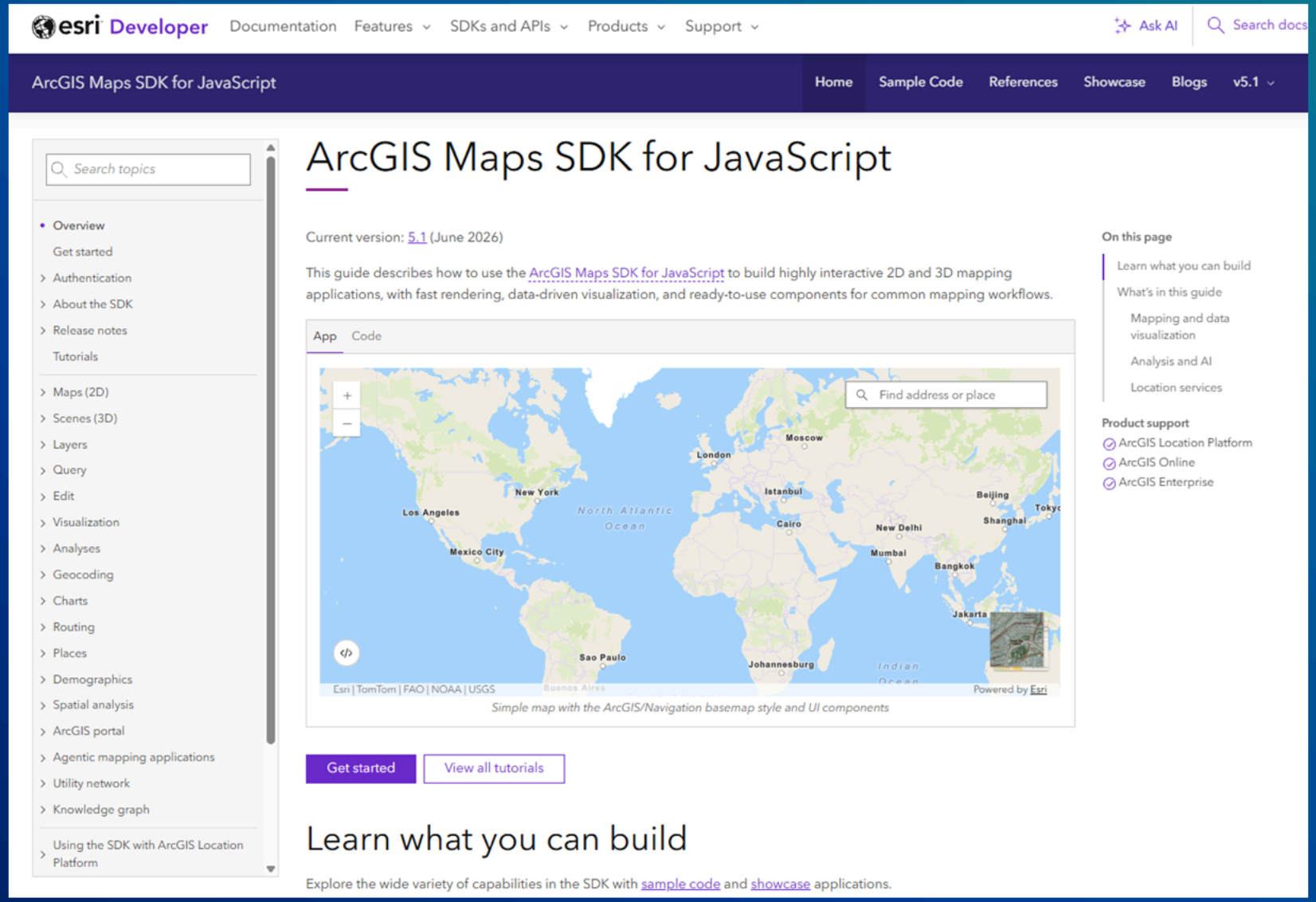
Summary

- Web Map and Web Scene
- Authoring and consumption across ArcGIS platform
- Dynamic maps via Popups, Arcade, Components and FeatureForms
- Authorizing apps via User or App authorization

References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Knowledge graph
Using the SDK with ArcGIS Location Platform
ArcGIS Maps SDK for JavaScript
Current version: 5.1 (June 2026)
This guide describes how to use the ArcGIS Maps SDK for JavaScript to build highly interactive 2D and 3D mapping applications, with fast rendering, data-driven visualization, and ready-to-use components for common mapping workflows.
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Find address or place
Moscow
London
New York
Istanbul
Beijing
Los Angeles
North Atlantic Ocean
Cairo
Shanghai
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Mexico City
Mumbai
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Jakarta
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- Collect your ticket with boat assignment from the Registration Desk from 10:00 – 16:30.
- Shuttles will begin loading after sessions conclude at 17:00.
- Boats will depart promptly at **18:15**, be sure to arrive early!

Questions

Acknowledgments

- Markstammdatenregister, Data License Germany – Attribution – Version 2.0
- Photos by Thomas Réaubourg on Unsplash

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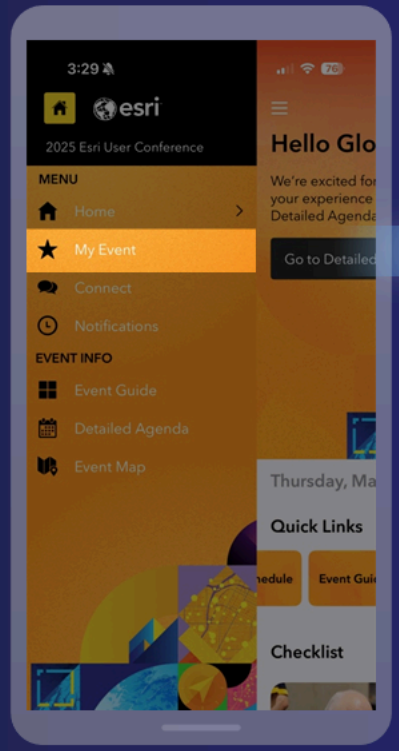

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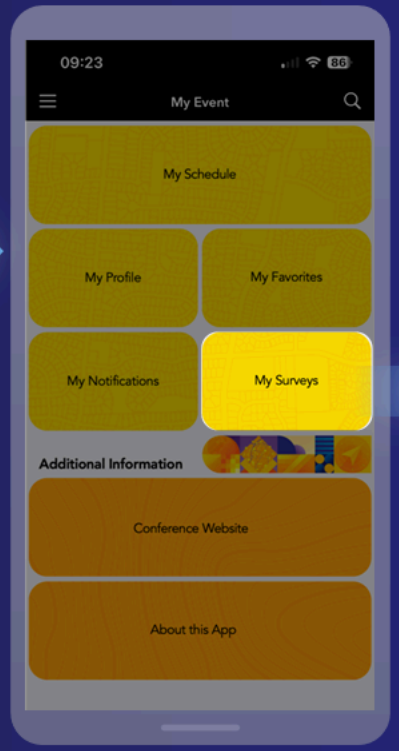
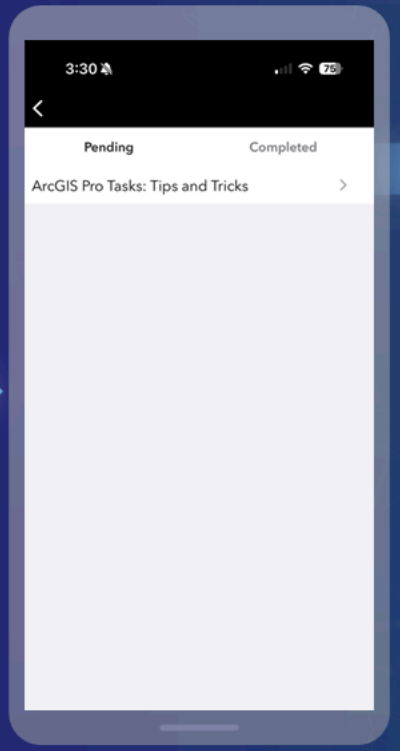
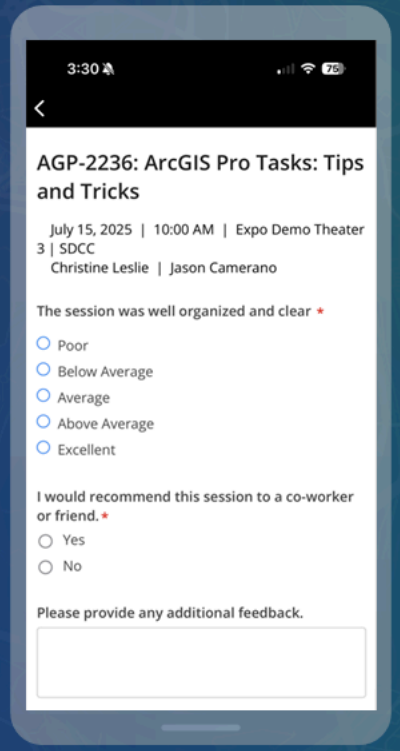


Select the desired survey



Complete the survey



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